SAYAN GHOSH

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Professional Summary

Highly motivated Computer Science student with expertise in **Artificial Intelligence** and **Java development**. Proven ability in practical skills like **Generative AI integration (Spring AI)**, Deep Learning deployment, and strong command of **Data Structures and Algorithms**. Seeking a challenging Software Engineering or AI/ML internship to apply technical aptitude and drive innovation.

Education

2023–2027 **Bachelor of Technology in Computer Science**, *VIT Bhopal University*, Bhopal, Madhya Pradesh, [7.65/10.0]
Expected Graduation: May 2027

Technical Skills

Programming	**Java**, Python, SQL, Bash
AI/ML/Data	Deep Learning, Machine Learning, TensorFlow/Keras, **OpenCV**, Scikit-learn, Pandas, Model Evaluation
Frameworks	**Spring AI**, **Spring Boot**, Streamlit, Git
Database/Cloud	PostgreSQL, MySQL, Cloud Architecture (Conceptual), Docker, Linux
Concepts/Tools	OOP, REST APIs, System Design, Data Structures & Algorithms (DSA)

Competitive Programming & Achievements

- Solved over **250+ LeetCode problems** (DSA, Algorithms, SQL), participated in **25+ Contests**, and achieved a **Codeforces rating of 1100**.

Key Coursework

- **Data Structures and Algorithms (DSA)**; Database Management Systems (DBMS); Computer Networks and Security
- Cloud Architecture and Deployment; Operating Systems; Artificial Intelligence / Machine Learning Fundamentals
- Object-Oriented Programming (Java)

Projects

- **1. Generative AI Chatbot Service** (Spring AI Focus)
  - Engineered a microservice using Spring Boot and the **Spring AI framework** for RAG-based knowledge retrieval.
  - Implemented RAG via a vector database, enabling the bot to answer complex queries from proprietary documentation.
  - Achieved an **85% accuracy rate** and demonstrated a reduction in manual lookup time by **40%**.
- **2. Image Captioning System** (Deep Learning & Attention)
  - Developed an image caption generation system using a **sequence-to-sequence architecture** (MobileNetV2 CNN + LSTM).
  - Integrated a **Neural Attention Mechanism** to significantly improve caption quality, trained on the Flickr8k dataset.
  - Deployed the full system as an interactive web application using **Streamlit**.
- **3. Interactive Color Detection Tool**
  - Developed an interactive utility using **Python/OpenCV** that detects and displays color names upon clicking an image pixel.
  - Employed a color-matching algorithm (Euclidean distance) using **Pandas** for efficient data mapping and identification.